



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



NASA Core Flight System Vulnerability Daily Space Systems Review Update

Threat Report

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AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

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NASA Core Flight System Vulnerability Daily Space Systems Review Update

1. Daily Update Scope

This 2026-08-13 automation run selected this item for daily analyst review based on current or recently updated public reporting. Where sources are older than 72 hours, the report explicitly treats the source window as expanded to seven days because the topic remains operationally relevant.

2. Executive Summary

CISA's NASA Core Flight System advisory remains relevant for space-system operators, labs, and contractors that integrate mission software into test or operational pipelines. Treat reachable test benches, ground systems, and CI/CD environments as exposure points even if flight safety systems are isolated. This report converts public reporting into immediate defender actions, detection logic, and threat-hunting queries. Claims are limited to the cited sources and Lost Boys Cyber analytic implications.

3. Intelligence Requirements

Question	Current Answer	Confidence
Why review this today?	It is current or recently updated and has practical exposure across monitored environments.	Medium
Which teams should act?	SOC, identity, endpoint, cloud/SaaS, vulnerability management, and business-system owners.	Medium
What evidence should be collected first?	Endpoint process/network telemetry, identity sign-ins, admin audit logs, proxy/DNS logs, and third-party access logs.	High

4. Source Review & Confidence

Source	Contribution	Confidence
CISA NASA Core Flight System advisory	Primary advisory for NASA cFS exposure.	Medium-High
NASA Core Flight System project	Product/project context.	Medium-High
CISA ICS defense-in-depth guidance	Mitigation context for mission/ICS-like systems.	Medium-High

5. Threat Activity Overview

CISA's NASA Core Flight System advisory remains relevant for space-system operators, labs, and contractors that integrate mission software into test or operational pipelines. Treat reachable test benches, ground systems, and CI/CD environments as exposure points even if flight safety systems are isolated.

Activity Element	Analyst Assessment	Collection Priority
Initial access	May involve exploitation, social engineering, identity abuse, or supplier	High

	compromise depending on environment.	
Execution / foothold	Look for script execution, unusual child processes, package hooks, API abuse, or living-off-the-land tools.	High
Command and control	Monitor rare direct-to-IP, unusual TLS destinations, SaaS/API anomalies, and abnormal OAuth/token flows.	Medium
Impact	Data theft, extortion, service disruption, or supply-chain propagation are plausible business impacts.	Medium

6. Affected Sectors and Exposure

Sector / Environment	Relevance	Priority Checks
Enterprise IT	Identity, endpoints, SaaS, and perimeter services are common attack surfaces.	MFA posture, EDR coverage, admin logs
Managed service / supply chain	Trusted relationships amplify blast radius.	Vendor access inventory, RMM restrictions, provenance
Critical operations	Outages may become operational disruptions even when OT is segmented.	BCP playbooks, segmentation, monitoring

7. Aviation and Aerospace Sector Impact

Segment	Operational Relevance	Exposure Path	Defender Action
Airlines / airports	Disruption can affect passenger processing, baggage, crew scheduling, and operations coordination.	Shared SaaS, identity providers, VPN, help desk, managed IT.	Validate third-party access and rehearse manual fallback.
Aerospace / defense manufacturing	IP theft and production disruption can affect design, supplier delivery, and maintenance schedules.	Engineering SaaS, supplier portals, RMM, VPN, code/package dependencies.	Segment engineering networks and monitor privileged vendor access.
Satellite / space / avionics support	Ground systems and telemetry pipelines can be high-impact even when flight safety systems are isolated.	Ground-segment IT, mission workstations, CI/CD, vendor appliances.	Review remote access, log retention, and mission-team escalation.
Aviation supply chain / MRO	Third-party compromise can cascade into airline and OEM operations.	Contractor identity, ticketing, document exchange, update mechanisms.	Inventory trust relationships and require breach notification playbooks.

8. Tactics, Techniques, and Procedures

Tactic / Phase	Observed or Plausible Technique	Evidence / Rationale
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Initial Access	T1190 Exploit Public-Facing Application / T1566 Phishing / T1195 Supply Chain Compromise	Current reporting emphasizes exploitation, social engineering, or package/supplier paths.
Execution	T1059 Command and Scripting Interpreter	Post-compromise actions often require scripts, installers, or living-off-the-land execution.
Persistence	T1098 Account Manipulation	Account/token changes are common in identity-heavy intrusions.
Command and Control	T1105 Ingress Tool Transfer / T1090 Proxy	Payload retrieval or proxying may follow initial foothold.
Impact	T1486 Data Encrypted for Impact / T1041 Exfiltration Over C2 Channel	Ransomware/extortion and data theft remain common outcomes.

9. Infrastructure and Indicators

Hard IOCs were not consistently available across all public sources reviewed for this daily item. The practical observable set is behavioral.

Observable	Type	Operational Use
Unexpected package install, script, or automation job	Behavioral	Supply-chain and execution triage
Unusual direct-to-IP outbound traffic	Network behavior	Malware C2 hunting
New OAuth grant, token, or admin session	Identity behavior	Account takeover triage
RMM/tooling execution outside maintenance window	Endpoint behavior	Intrusion/ransomware precursor

10. Detection Engineering

10.1. Detection Summary

Signal	Telemetry Source	Analytic Goal
Suspicious tool/package execution	DeviceProcessEvents / Sysmon	Find initial execution and staging
Direct-to-IP or rare outbound connections	DeviceNetworkEvents / firewall	Find C2 or staging without DNS
Identity and SaaS access anomalies	SigninLogs / audit logs	Find trusted-relationship abuse
<pre> title: NASA Core Flight System Vulnerability Daily Space Systems Review Update Daily Priority Signals id: nasa-core-flight-system-vulnerability-daily-space-systems-review-update-20260813 status: experimental logsource: product: windows category: process_creation detection: selection_terms: CommandLine contains: 'nasa' CommandLine contains: 'cfs' CommandLine contains: 'core flight' CommandLine contains: 'mission' CommandLine contains: 'space' condition: selection_terms </pre>		

level: medium

11. Threat Hunting

11.1. Hunt Hypothesis

If this activity is present, analysts should find one or more of: suspicious execution tied to the topic keywords, outbound communications to rare infrastructure, or identity events inconsistent with normal administrative patterns.

```
let lookback = 14d;
DeviceProcessEvents
| where Timestamp > ago(lookback)
| where ProcessCommandLine has_any ('nasa', 'cfs', 'core flight', 'mission', 'space') or FileName
has_any ('nasa', 'cfs', 'core flight', 'mission', 'space')
| project Timestamp, DeviceName, AccountName, FileName, ProcessCommandLine, InitiatingProcessFileName
| union (DeviceNetworkEvents
| where Timestamp > ago(lookback)
| where RemoteUrl has_any ('nasa', 'cfs', 'core flight', 'mission', 'space') or
InitiatingProcessCommandLine has_any ('nasa', 'cfs', 'core flight', 'mission', 'space') or RemoteIPType
== 'Public'
| project Timestamp, DeviceName, AccountName=InitiatingProcessAccountName,
FileName=InitiatingProcessFileName, ProcessCommandLine=strcat(RemoteUrl, ':', RemotePort),
InitiatingProcessFileName)
SigninLogs
| where TimeGenerated > ago(14d)
| where AppDisplayName has_any ('nasa', 'cfs', 'core flight', 'mission', 'space') or UserAgent has_any
('nasa', 'cfs', 'core flight', 'mission', 'space')
| summarize count(), makeset(IPAddress, 10), makeset(AppDisplayName, 10) by UserPrincipalName,
bin(TimeGenerated, 1d)
```

12. Risk Assessment

Risk Dimension	Assessment	Rationale
Likelihood	Medium	Topic is current and broadly applicable, but local exposure varies.
Impact	High	Identity, supply-chain, extortion, or operational dependencies can create material disruption.
Detection Confidence	Medium	Behavioral analytics require tuning and source-specific enrichment.

13. Recommended Actions

Priority	Owner	Action
Critical	SOC	Run the included hunts for the last 14 days and review high-confidence hits.
High	Identity / SaaS	Review OAuth grants, privileged sessions, and anomalous admin changes.
High	Endpoint / Network	Monitor rare direct-to-IP egress, script execution, and RMM/tool misuse.
Medium	Business Owner	Validate continuity and third-party escalation paths for impacted workflows.

14. References

- [1] CISA NASA Core Flight System advisory: <https://www.cisa.gov/news-events/ics-advisories/icsa-26-197-03>
- [2] NASA Core Flight System project: <https://github.com/nasa/cFS>
- [3] CISA ICS defense-in-depth guidance: <https://www.cisa.gov/resources-tools/resources/improving-industrial-control-systems-cybersecurity-defense-depth-strategies>